A Kalman filter benchmark for data-driven forecasts of GRACE terrestrial water storage

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Abstract. Data-driven forecasts of terrestrial water storage are usually scored by their agreement with the observations, or against competing models, rarely against the storage record they start from. How much skill that record alone supplies is unclear. Persistence, the simplest such reference, carries forward the latest deseasonalized residual, including whatever transient part it contains. We test whether filtering the Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On (GRACE-FO) record before extrapolation gives a stronger reference. For each basin we fit a Kalman filter with a first-order autoregressive state and additive model noise, and forecast from the filtered state. The split between state and noise is statistical and does not separate physical variability from observation error. Forecasts are evaluated on deseasonalized residuals over 234 basins and five expanding-window test periods in 2019–2026. On the Center for Space Research (CSR) mascon product the filter reduces mean squared error by 5.0 % at a 1-month lead and 8.8 % at 2 months, against the stronger of two damped-persistence references. Refitting with zero model-noise variance removes that 1-month gain. On the Jet Propulsion Laboratory (JPL) product the filter improves on damped persistence at 2–6 months; the 1-month difference is not significant. On CSR, a ridge regression on 12 months of filtered storage and ERA5-Land variables lowers standardized error by a further 7.6 % at 1 month, although its pooled error in centimeters is higher than the filter's. No nonlinear model tested has lower error than this ridge at 1–3 months. Both the filter and damped persistence have lower standardized error than the GRACE-FCast member built on the same mascon product, on the fully covered basins, at 1 month. GRACE-FCast has significantly lower error than the filter from 3 to 5 months onward, depending on product and variant. Storage forecasts should be scored against a filtered reference and in physical units.

1 Introduction

The Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On (GRACE-FO) missions have measured monthly changes in terrestrial water storage (TWS) since 2002, integrating changes in groundwater, soil moisture, surface water, snow and ice (Tapley et al., 2004; Landerer et al., 2020). Anomalies of this quantity (TWSA) have a memory of several months, which makes them both a forecast target and a source of predictability. TWSA forecasts support drought early warning and water management (Li et al., 2026; Arsenault et al., 2020), flood-potential assessment (Reager et al., 2014), and the bridging of the two- to three-month latency of GRACE-FO data releases (Mo et al., 2025).

A growing number of studies forecast TWSA directly from data. Neural networks have been trained on GRACE and hydrometeorological series at basin scale (Ahmed et al., 2019), at grid scale globally (Li et al., 2020, 2024; Li and Kusche, 2026), for Europe with subsequent assimilation (Li et al., 2025), and for latency filling (Mo et al., 2025). Statistical time-series models have been applied per basin (Ahi and Cekim, 2021), benchmark predictability has been quantified at decadal scale (Zhu et al., 2019), and physically based systems forecast TWS with seasonal meteorological forcing (Li et al., 2026; Getirana et al., 2020). Evaluation differs among these studies. Correlation measures agreement with observations without specifying a reference forecast, whereas Nash–Sutcliffe efficiency uses the observed mean as its reference. Li et al. (2024) evaluate forecasts against GRACE observations, and Li and Kusche (2026) compare with SEAS5. Such comparisons do not establish how much a model improves on the recent storage history. Damped persistence provides a direct test of that improvement, and of the ten data-driven TWSA forecasting studies we examined, none reports it. At sub-seasonal to seasonal leads, the FEWS NET land data assimilation hindcast of Li et al. (2026) concludes that initial conditions dominate sub-seasonal to seasonal TWS skill, but measures that dominance with a lag-autocorrelation diagnostic rather than with a benchmark forecast.

We test whether filtering the storage record gives a stronger benchmark. Basin-average TWSA is subject to observation noise (Save et al., 2016; Landerer et al., 2020). We work with the residual after removing a fitted trend and seasonal cycle. A persistence forecast carries forward the latest residual, including any transient part of that month's value, unchanged for raw persistence and attenuated for damped persistence. The benchmark we propose is a per-basin Kalman filter (Kalman, 1960) with a first-order autoregressive (AR(1)) state and an additive noise term in the observation equation, which we call model noise. Section 3.4 defines this term. It is a statistical component of the residual that may mix fast hydrological variability, errors in the trend and seasonal fit, and instrument and processing errors; nothing in the method separates these contributions or measures the accuracy of the satellite data. The filter estimates a persistent state from the residual before extrapolation, has three parameters per basin, and needs no data beyond the storage series itself. Because it propagates its state through months without an observation, it also remains defined across the eleven-month gap between GRACE and GRACE-FO without interpolation. Its treatment of missing observations follows from the state-space model; forecasts issued within that gap are not evaluated here.

The same argument has been made in sea-ice forecasting, where Niraula and Goessling (2021) showed that damped-persistence references that "do not exploit observations optimally" overstate the skill of dynamical forecast systems, and built a stronger reference from spatial damping. Here we follow the same reasoning for satellite-observed TWSA, and strengthen the reference by filtering the storage record. Kalman methods are well established in the GRACE literature, although they have been used for other purposes. For example, ensemble Kalman filters assimilate TWSA into land-surface models (Springer et al., 2026), forward Kalman filters predict gravity-field coefficients (Kankanige et al., 2026), and integrated postprocessing separates the non-seasonal signal from noise in the monthly solutions (Zhang et al., 2025). None of these uses a univariate filter on the finished basin series as a benchmark forecast. A benchmark for TWS prediction has been proposed before by Zhu et al. (2019), who quantified decadal TWS predictability with an elasticity framework, whereas our benchmark addresses monthly lead times. Most closely related to our work, Nie et al. (2025) showed on 515 basins that deep architectures do not beat linear regression for TWS prediction when autoregressive information is excluded from the feature set by design. Their design has

no lagged storage, so it has neither a persistence nor a filter benchmark. Our results complement theirs: when lagged storage is included, the outcome of the model comparison depends more on how the storage record is filtered than on the choice of architecture.

We compare the filter with persistence and conventional statistical forecasts, then use ablation experiments to assess the contribution of its model-noise term. We also test whether a basin's forcing history improves the filtered forecast and whether nonlinear models outperform linear regression on the same inputs. Finally, we compare with the published GRACE-FCast hindcast (Li and Kusche, 2026) using two independently processed mascon products. Section 2 describes the data, Sect. 3 the forecast target and the models, Sect. 4 the results and Sect. 5 their scope and limits.

2 Data

2.1 GRACE and GRACE-FO mascon solutions

The primary storage data set is the Center for Space Research (CSR) RL0603M mascon solution (Save et al., 2016; Save, 2020) on a $0.25°$ grid, April 2002 through May 2026, in centimeters of equivalent water height; the product version is RL06.2. Each monthly solution is assigned to its official calendar month using the product's documented list of missing months rather than by the midpoint of its data span, which would mislabel two solutions. The second storage data set is the Jet Propulsion Laboratory (JPL) RL06.3Mv04 mascon solution (Wiese et al., 2023) with the coastline resolution improvement filter applied, sampled at $0.5°$, April 2002 through June 2026, with the product's scale factors applied and its official missing-month metadata used. The products have different native estimation geometries. CSR uses equal-area cells approximately $1°$ wide at the equator, whereas JPL uses approximately $3°$ mascons (Center for Space Research, 2026; Wiese et al., 2023). These cell sizes and the finer distribution grids do not represent the effective spatial resolution of the observations. CSR controls correlated errors through regularized estimation without additional empirical smoothing or destriping (Center for Space Research, 2026). We apply no further spatial filter to either product. The two solutions are independently processed estimates from the same satellite observations; comparing them tests sensitivity to processing choices rather than providing independent observational replication.

2.2 Basins

Basins follow a HydroSHEDS-based Level-3 basin and mascon partition (Lehner and Grill, 2013). Each basin series is the cosine-latitude area-weighted mean of the grid cells in its mask. Of the 284 basins in the partition we exclude 46 ice-sheet basins in Antarctica and Greenland and 4 water bodies, leaving 234 basins on the CSR product. On the JPL product six small-island masks contain no valid scaled cells, so 228 basins are kept there. Figure 1 maps the basins and marks the subsets used in the comparison with GRACE-FCast (Sect. 3.6).

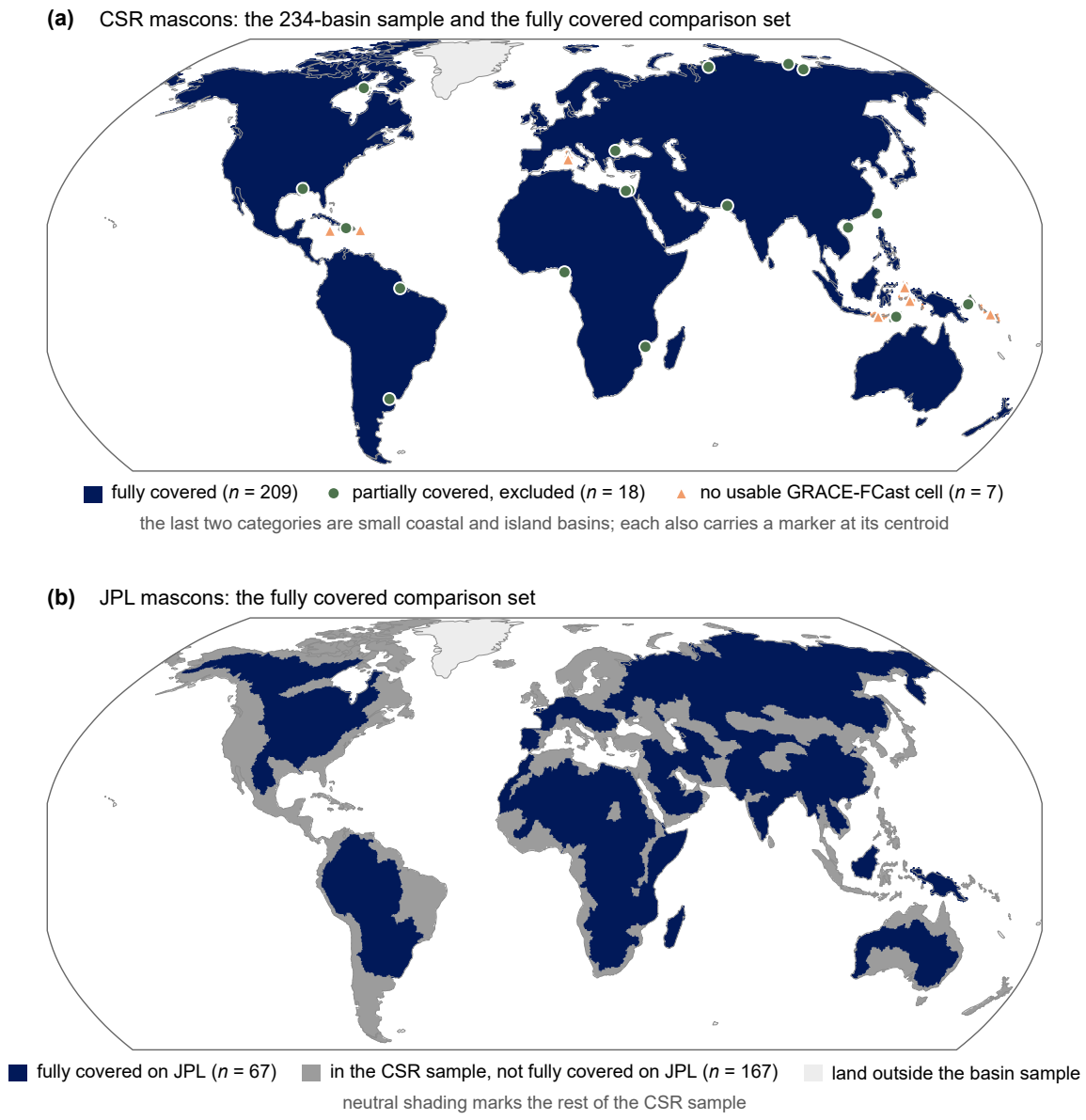


Figure 1. The basin sample. (a) The 234 basins kept on the CSR product, colored by their role in the comparison with GRACE-FCast (Sect. 3.6): 209 basins are fully covered by both the mascon grid and the forecast grid and enter the comparison, 18 have partial coverage and are excluded by the coverage rule, and 7 have no usable GRACE-FCast cell. (b) The JPL product: the 67 basins that pass the coverage rule there, fewer than on CSR because JPL's coarser native mascons (Sect. 2.1) are fully contained by fewer basins. The remaining basins of the CSR sample are shaded neutral; the three-way split of (a) was not evaluated on JPL. Land outside the basin sample is pale grey in both panels. Basin partition based on HydroSHEDS (Lehner and Grill, 2013); base map made with Natural Earth (public domain). Colour palettes use scientific colour maps (Crameri et al., 2020).

2.3 ERA5-Land forcing and climate indices

We use 11 ERA5-Land variables (Muñoz-Sabater et al., 2021) at $0.5°$, regridded server-side from $0.1°$, for January 2001 through May 2026: total precipitation, total evaporation, runoff (total, surface and subsurface), volumetric soil water in four layers, 2-m temperature, and snow depth water equivalent. Accumulated fluxes are converted from mean daily rates to calendar-month totals, the same months to which the storage solutions are assigned (Sect. 2.1). Evaporation is sign-flipped so that positive values mean moisture loss, temperature is converted to degrees Celsius, and snow depth to centimeters of water equivalent. Basin aggregation uses the same cosine-latitude weighted means as the storage aggregation, mapping each $0.25°$ mask cell to its nearest $0.5°$ ERA5 cell. Because ERA5-Land is defined over land only, coastal and island basins aggregate their forcing from the land fraction of their mask. Of the 234 basins, 79 draw less than 90 % of their mask weight from valid ERA5 land cells and 13 less than 50 % (minimum 24.5 %). No basin is excluded on this ground. Twelve monthly climate indices from NOAA PSL enter one benchmark model only (Sect. 3.3). Two of them, AMM and PDO, end before the test window and are dropped from it.

2.4 GRACE-FCast

For the external comparison we use the public GRACE-FCast hindcast of Li and Kusche (2026), downloaded from PANGAEA (Li, 2025). It is released as three members built on the JPL, CSR and GSFC mascons. We use the CSR member against our CSR target and the JPL member against our JPL target, so that each comparison shares a gravity solution. The CSR member holds monthly $1°$ global forecasts at lead times of 1 to 12 months, initialized December 2009 through May 2024 (174 initializations), and the JPL member holds 173 initializations through April 2024. GRACE-FCast forecasts the interannual and sub-seasonal components of TWSA with separate long short-term memory networks. Their inputs are antecedent hydrometeorological and oceanographic predictors, each selected by correlation from a range of lags and neighboring locations; lagged storage is not among them. The full forecast adds seasonal and trend components extrapolated from the previous year of GRACE observations (Li and Kusche, 2026). We evaluate both the full product and its non-seasonal component (Sect. 3.6).

Li and Kusche (2026) validate the average of the three forecast members against the average of the three mascon solutions; our results apply to the individual members scored here. Their JPL member was trained against JPL RL06.1, whereas our JPL target is RL06.3Mv04, so the JPL comparison involves two different releases. Their training also fills the 2017–2018 mission gap with the reconstruction of Yin et al. (2023), which is not used by any of our models. Their networks are retrained, and their predictors reselected, at each monthly initialization (Li and Kusche, 2026), whereas our fitted parameters are held fixed within each evaluation fold (Sect. 3.2).

3 Methods

3.1 Forecast target

For each basin and each evaluation fold we fit, on the training window only, a climatology consisting of a linear trend plus annual and semiannual harmonics (six coefficients, ordinary least squares). The frozen coefficients are used to deseasonalize both the training and the test data, and the residual is divided by its training-window standard deviation. The forecast target is this deseasonalized, per-basin standardized residual. It retains the interannual and shorter-term variations relevant to drought and flood applications. Standardization expresses each basin's error relative to its own training variability; it does not impose equal realized error contributions. Section 4.5 and Appendix C report the same forecasts in centimeters, where basins with larger storage variability contribute more to pooled error.

3.2 Evaluation design

We use five expanding-window folds. Their issue windows begin in June 2019, November 2020, April 2022, September 2023 and February 2025, and together cover June 2019 through April 2026 at a 1-month lead, with targets reaching May 2026. Fold membership is defined by the forecast issue date. Within each fold, every climatology, filter parameter, model coefficient and standardization constant is fit only on observations before the fold's first issue date, and every test forecast is issued at or after that date. A forecast issued before the fold start but targeting a month after it belongs to neither set. Defining membership by target date instead would let a lead-h forecast use transforms fit up to $h-1$ months after its issue date. Earlier windows are excluded from this common-sample evaluation because the required lagged predictors are incomplete across the 2017–2018 gap. Every model is scored on the same rows at a given lead time, so the sample on the CSR product is $n = 19{,}422$, $19{,}188$, $18{,}954$, $18{,}720$, $18{,}486$ and $18{,}252$ basin-months at lead times of 1 to 6 months. On the JPL product, whose record runs one month longer, the 228 basins carry 84 issue months at a 1-month lead against 83 on CSR, so $n = 19{,}152$ there. Issue time is defined relative to the last available observation, so the release latency of GRACE-FO is not modeled and all results are retrospective potential skill.

3.3 Benchmark forecasts

We evaluate the following benchmark forecasts:

- climatology, which forecasts a residual of zero;
- persistence, the last observed residual carried forward unchanged;
- damped persistence in two variants: the ρ^h variant multiplies the last residual by a per-basin AR(1) coefficient ρ raised to the power of the lead time, and the regression variant regresses the lead-h target directly on the last residual;
- a pooled ridge regression on own-basin lagged residuals (penalty 1; lags 0, 1, 2, 5 and 11 months; one model for all basins), and the same ridge with the lagged climate indices of Sect. 2.3 added;

– a per-basin ridge on the same lags, refit separately for each basin.

For the comparison reference, we select the damped-persistence variant with the lower pooled test RMSE at each lead time: on CSR the ρ^h variant at 1 month and the regression variant at 2 to 6 months, on JPL the ρ^h variant at 1 and 2 months and the regression variant at 3 to 6 months (Appendix B). This retrospective choice provides the more demanding of the two references; it is not a selection rule available at forecast issue time. The weaker variant has up to 15 % higher mean squared error. Each table and figure states the reference used.

3.4 The Kalman filter benchmark

The proposed benchmark models each basin's deseasonalized residual y_t as the sum of an AR(1) latent state x_t and a white model-noise term η_t:

$$x_t = \rho\, x_{t-1} + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, q), \tag{1}$$

$$y_t = x_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, r). \tag{2}$$

We call η_t the model noise; r is its variance and q the process-noise variance. The latent state x_t is a statistical estimate of the persistent part of the residual. The three parameters (ρ, q, r) are estimated per basin and fold by maximum likelihood on training months only, using L-BFGS-B from several starting points. The filter runs forward through the training and test months and updates its state estimate $x_{t|t}$ at each observed month with the Kalman gain, which depends on ρ and on the ratio q/r. At a month with no observation, including the eleven months of the mission gap, the filter skips the update and propagates its state and state uncertainty forward, so that no interpolation or gap filling is required. The forecast for lead time h issued at month t is

$$\hat{y}_{t+h|t} = \rho^h\, x_{t|t}, \tag{3}$$

which applies damping to the filtered state. We refer to this forecast as the Kalman filter (KF). The state-space model contains one scalar storage state per basin and no hydrological model.

We use two variants of the filter to explain its margin over damped persistence. In the *zero-r variant* the same model is refit with the same likelihood and with r fixed at zero. This forces the Kalman gain to one, so the state tracks the observed residual and the forecast reduces to damped persistence with a maximum-likelihood ρ. The comparison measures the effect of allowing a nonzero model-noise variance, including the resulting changes in the fitted state parameters. In the *two-variance variant* the observation equation has separate model-noise variances for months before January 2018 and for GRACE-FO months from then on, so the parameter vector is $(\rho, q, r_{\mathrm{GRACE}}, r_{\mathrm{FO}})$. We require at least 12 observed GRACE-FO months in the training window to fit a separate variance; otherwise, the fit uses the single-variance model.

The likelihood determines the allocation of residual variability between the evolving state and the additive model-noise term. The model-noise term is presumed to be a mixture of fast hydrological variability, errors in the fitted trend and seasonal cycle, and instrument or processing errors; their shares are not estimated. Unpredictable changes can also enter the state through the

process-noise term, so r does not represent all variation that cannot be forecast. In 259 of the 1170 basin-fold fits the likelihood drives r to effectively zero (below 10^{-6} in standardized units). The fitted r is a model parameter, not a measure of observation error in the mascon solutions.

3.5 Forecast models built on the filter

We call the following models own-basin models because each uses only its own basin's storage record and forcing. Every one of them predicts the error of the Kalman filter forecast, and its own forecast is the filter forecast plus that predicted error. Table 1 lists the models and the abbreviations used for them throughout. In the abbreviations, R denotes a ridge correction, the number the months of filtered-state history it uses, and E the ERA5-Land forcing. Four models are linear. KF-R1 regresses the error on the current filtered state alone. KF-R12 regresses it on the last twelve months of the filtered state. KF-R12E adds the twelve months of all 11 ERA5-Land anomaly series, giving 12 by 12 features. KF-R1E instead adds the ERA5-Land anomalies at lags of 0, 1 and 2 months to the current filtered state. Each ERA5-Land basin series is deseasonalized with the same train-window trend and harmonic fit as the storage series (Sect. 3.1), standardized by its training-window standard deviation and clipped at ± 10 standard deviations before it enters any model. All four ridges are pooled across basins with penalty 1 on train-standardized features. Three models are nonlinear and use the same information. The LSTM is a shared-encoder long short-term memory network (Hochreiter and Schmidhuber, 1997) over the same twelve-month window of filtered state and the 11 ERA5-Land channels. It is trained with a time-ordered split that holds out the last 15 % of training months for early stopping, and its two seeds are reported as a seed ensemble. The MLP is a multilayer perceptron (hidden layers of 64 and 32 units, early stopping) on the KF-R12E feature vector, with two seeds. The residual MLP is a two-stage model, KF-R1E followed by an MLP of the same size trained on the first stage's training residual, with three seeds. Finally, two neighbor models, not adopted (Sect. 4.6), add to KF-R1 the filtered state of one other basin propagated to the target month, with that basin chosen by correlation or by distance. Five of these forecasts are also reported in centimeters (Sect. 4.5 and Appendix C): the stronger damped-persistence variant, KF, KF-R1, KF-R12 and KF-R12E.

3.6 Comparison with GRACE-FCast

The GRACE-FCast hindcast (Li, 2025) is aggregated to our basin geometry with exact $0.25°$ to $1°$ cell-containment weights. Its forecasts are moved into each fold's standardized residual space as follows. The full product has our train-window climatology subtracted; the non-seasonal component, their interannual plus sub-seasonal forecast, does not. Both then receive a per-fold, per-basin, per-lead mean offset estimated on pre-test data only, which requires at least 24 such months, and are divided by our training-window residual standard deviation. This aligns their mean levels with our target; the two decomposition conventions otherwise differ. Scoring both variants shows how their extrapolated seasonal and trend terms affect error against our target. Every model is scored on the identical set of basin-months at every lead time. On CSR it spans 60 test months: folds 1 to 3 in full, 9 of the 17 issue months of fold 4, and none of fold 5, because GRACE-FCast's final initialization, May 2024, falls inside fold 4. On JPL, whose final initialization is one month earlier, it spans 59 months.

Table 1. Forecasts evaluated in this paper, with the section that defines each. Appendix A gives the identifier of each in the code and data repository.

Name	Definition
Climatology	Zero-residual forecast (Sect. 3.3).
Persistence	Last observed residual carried forward (Sect. 3.3).
Damped persistence	Last residual damped toward zero; ρ^h and regression variants, always scored against the stronger at each lead time (Sect. 3.3).
Pooled lag ridge	Ridge regression on own-basin lagged residuals, one model for all basins; the index variant adds lagged climate indices (Sect. 3.3).
Per-basin lag ridge	The pooled lag ridge's features, refit separately for each basin (Sect. 3.3).
KF	The proposed benchmark: damped persistence of the filtered state from the per-basin AR(1) plus model-noise specification (Sect. 3.4).
KF, $r = 0$	The zero-r variant, refit with the model-noise variance fixed at zero (Sect. 3.4).
KF, two variances	The two-variance variant, with separate model-noise variances for GRACE and GRACE-FO months (Sect. 3.4).
KF-R1	KF plus a ridge correction on the current filtered state (Sect. 3.5).
KF-R12	KF plus a ridge correction on the 12-month history of the filtered state (Sect. 3.5).
KF-R12E	KF-R12 with the 12-month history of the 11 ERA5-Land anomalies added (Sect. 3.5).
KF-R1E	KF plus a ridge correction on the current filtered state and ERA5-Land anomalies at lags of 0 to 2 months (Sect. 3.5).
LSTM	Shared-encoder LSTM over the 12-month history of filtered state and ERA5-Land channels; two-seed ensemble (Sect. 3.5).
MLP	Multilayer perceptron on the KF-R12E features; two seeds (Sect. 3.5).
Residual MLP	KF-R1E followed by an MLP on its training residual; three seeds (Sect. 3.5).
Neighbor ridge	KF-R1 plus one other basin's propagated filtered state, selected by correlation or by distance (Sect. 3.5; not adopted, Sect. 4.6).
GRACE-FCast, full and non-seasonal	The published forecast of Li and Kusche (2026), with and without its extrapolated seasonal and trend terms (Sects. 2.4 and 3.6).

We define two basin sets. The matched set keeps every basin with at least one usable GRACE-FCast cell, which is 227 of the 234 basins on CSR and 227 of 228 on JPL. The fully covered set applies one rule to both products: a basin enters only if it fully contains every 0.25° mask cell of at least one native mascon of the product being scored, and all sixteen cells of at least one finite 1° GRACE-FCast cell. This rule tests cell containment directly and does not rely on an area or coverage threshold. On CSR, 229 basins contain a full native mascon, 211 contain a full forecast cell, and 209 satisfy both. On JPL, whose native mascons are coarser on the sampled grid, 67 do. The rule screens grid geometry only: a single native mascon is not an independently resolved gravity signal, because the effective resolution of the observations is coarser than the mascon and the regularized estimation couples neighboring mascons (Sect. 2.1). We report the fully covered set. On the matched set, the standardized 1-month error of the full product on JPL is 5.47 against 0.93 on the fully covered set. The gap arises because our standardization uses the residual standard deviation of our own product, which is small in basins that the coarser forecast grid only partly covers. Values on the matched set are therefore dominated by grid mismatch, and we do not interpret them as forecast accuracy.

3.7 Verification statistics

The primary metric is the pooled root-mean-square error (RMSE) in standardized units. We report it as the mean-squared-error skill score of a forecast A against a stated benchmark B,

$$S_{A|B} = 1 - \frac{\mathrm{MSE}_A}{\mathrm{MSE}_B}, \tag{4}$$

in percent, so that a positive value means A has the lower error. The forecast and reference are stated for each comparison. Pooled comparisons use the Diebold–Mariano test (Diebold and Mariano, 1995) on the monthly cross-basin mean loss differential, with the small-sample correction of Harvey et al. (1997) and a Newey–West variance at maximum lag $\max(h-1,1)$ (Newey and West, 1987). Confidence intervals use a moving-block bootstrap (Künsch, 1989) of the monthly losses with block length $\max(3,h)$ and 2000 draws, so that the block covers the lag-$(h-1)$ dependence of overlapping h-step forecasts. Per-basin significance uses per-basin Diebold–Mariano statistics on the scored test months with Benjamini–Hochberg false-discovery-rate control at $q = 0.10$ (Benjamini and Hochberg, 1995), following the field-significance reasoning of Wilks (2016). Basins that are helped and basins that are hurt are always counted separately. In the text we state whether a difference is significant at $p < 0.05$. Tables report two-sided p values, with values below 0.001 written as < 0.001. A small p value indicates a difference in mean loss; the sign of the skill score identifies which forecast has lower error. Pooled tests are not adjusted across model and lead-time comparisons; false-discovery-rate correction applies only to the stated families of per-basin tests.

4 Results

4.1 Benchmark forecasts

Table 2 and Fig. 2a compare the benchmark forecasts on the CSR product. KF has lower error than the stronger damped-persistence variant at every lead time, by 5.0 % at 1 month, 8.8 % at 2 months and 2.5 to 5.6 % at 3 to 6 months ($p < 0.001$ at

every lead time; Appendix B). The margin peaks at 2 months, and the 95 % bootstrap interval on the 1-month margin is $+3.9$ to $+6.1$ %. KF also improves on the per-basin lag ridge, the strongest conventional statistical benchmark at short lead times, by 1.0 to 3.9 %. The difference is significant ($p < 0.02$) at five of the six lead times. At 4 months it is not significant ($p = 0.078$), although the bootstrap interval excludes zero ($+0.02$ to $+2.20$ %). Against the pooled lag ridge KF is ahead by 3.9 and 4.2 % at 1 and 2 months ($p < 0.001$), with no significant difference at 3 to 6 months ($p \geq 0.18$). The advantage over this pooled model is restricted to short leads.

Table 2. Benchmark forecasts on the CSR product: pooled skill (%) against the stronger damped-persistence variant at each lead time h (months), on the same basin-months for every forecast (234 basins, five folds; n as in Sect. 3.2). Positive is better. The reference is the ρ^h variant at $h = 1$ and the regression variant at $h = 2$ to 6. The best value in each column is in bold. Appendix B gives the standardized RMSE and the Diebold–Mariano p value for every row.

Forecast	h=1	h=2	h=3	h=4	h=5	h=6
Climatology	-125.6	-70.7	-58.9	-51.4	-42.7	-33.6
Persistence	-20.7	-22.7	-21.4	-19.5	-16.0	-15.8
Damped persistence, weaker variant	-1.3	-0.6	-6.4	-11.8	-12.0	-10.0
Damped persistence, stronger variant	0.0	0.0	0.0	0.0	0.0	0.0
Pooled lag ridge	$+1.2$	$+4.8$	$+5.1$	$+3.6$	$+3.2$	$+2.5$
Pooled lag ridge, index variant	$+1.0$	$+4.5$	$+5.1$	$+3.7$	$+3.2$	$+2.4$
Per-basin lag ridge	$+3.1$	$+5.9$	$+4.3$	$+2.1$	$+0.4$	-0.3
KF	$+5.0$	$+8.8$	$+5.6$	$+3.1$	$+2.5$	$+3.6$
KF-R1	$+4.8$	$+7.9$	$+6.0$	$+5.4$	$\mathbf{+5.1}$	$\mathbf{+5.2}$
KF-R12	$+5.7$	$+10.1$	$+7.3$	$+4.8$	$+4.0$	$+4.2$
KF-R12E	$\mathbf{+12.2}$	$\mathbf{+13.7}$	$\mathbf{+9.6}$	$\mathbf{+6.2}$	$+4.6$	$+4.5$

Figure 3 shows the 1-month forecasts in four basins. In the Saharan and Algerian Atlas and in the East Nile Delta, where the month-to-month scatter is large relative to the slower residual variations, damped persistence follows each excursion of the last observation while the filter gives less weight to each new observation. In the Danube the two forecasts are close, and in the Amazon, where residual variability is large and smooth, they are nearly indistinguishable and damped persistence is marginally better. Across the sample, KF has the lower 1-month RMSE in centimeters in 158 of the 234 basins and damped persistence in 76. The gain is largest where the residual has the least month-to-month persistence, and in small and dry basins (Fig. 4c). Across the 234 basins the per-basin skill has a Spearman rank correlation of -0.34 with the lag-1 autocorrelation of the residual, -0.27 with basin area and -0.30 with mean annual ERA5-Land precipitation ($p < 0.001$ for each). In the driest quarter of the basins the median gain is 6.0 %, against 0.9 % elsewhere. The two largest gains are in the Egyptian Western Desert and the Saharan and Algerian Atlas (Fig. 3a).

Raw persistence has higher mean squared error than damped persistence by 16 to 23 %. Relative to damped persistence, KF reduces mean squared error by 5.0 to 8.8 % at 1 and 2 months. The per-basin lag ridge has higher error than KF at every lead

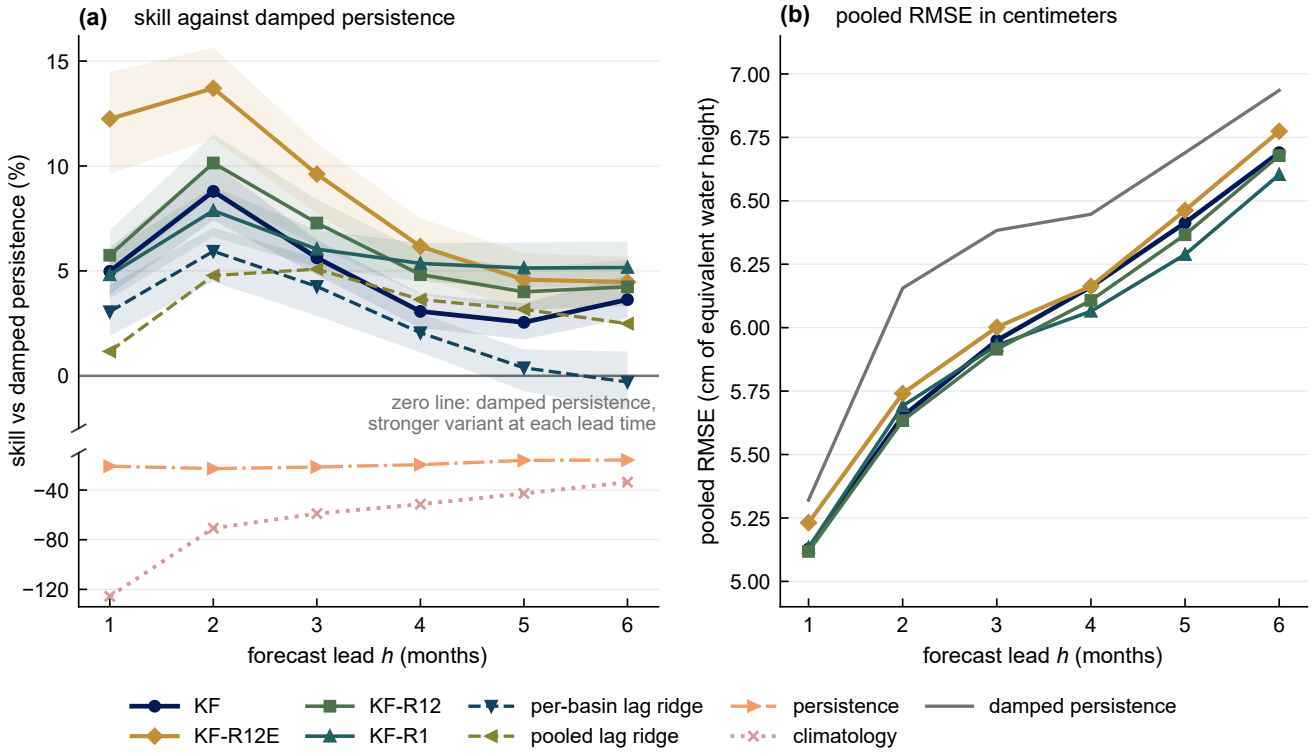


Figure 2. Benchmark forecasts on the CSR product. (a) Pooled skill (%) against the stronger damped-persistence variant (the ρ^h variant at a 1-month lead time and the regression variant at 2 to 6 months) for KF-R12E, KF, KF-R12, KF-R1, the per-basin lag ridge, the pooled lag ridge, persistence and climatology, the last two on a broken axis. Shading gives moving-block bootstrap 95 % confidence intervals for KF-R12E, KF, KF-R12, KF-R1 and the per-basin lag ridge. The gap between the zero line and KF is the gain from filtering the most recent observation before damping it. (b) Pooled RMSE in centimeters of equivalent water height by lead time for the five forecasts also reported in centimeters: damped persistence (stronger variant), KF, KF-R1, KF-R12 and KF-R12E. In this weighting the high-variance basins contribute more to pooled error and the ranking changes (Sect. 4.5).

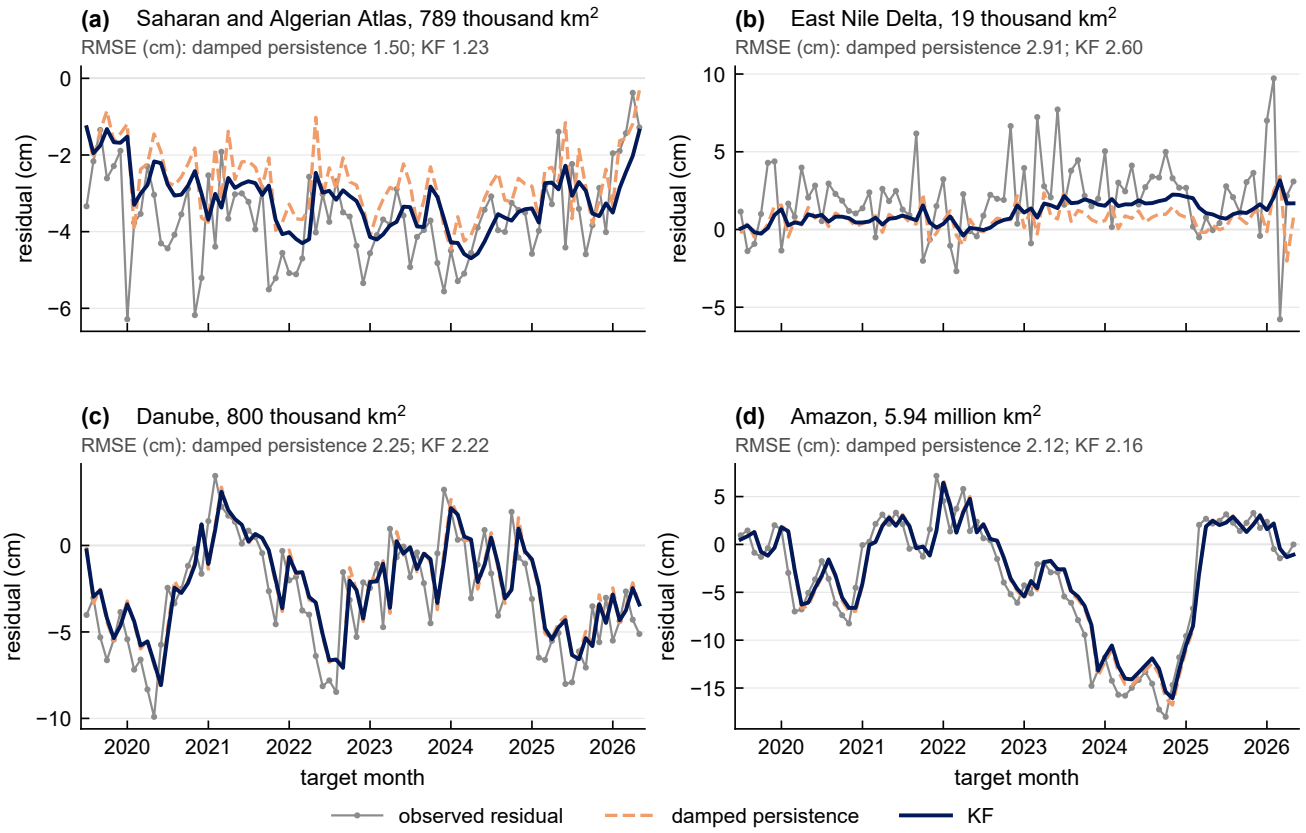


Figure 3. Observed deseasonalized storage residual (grey) and the 1-month forecasts of damped persistence (ρ^h variant, dashed) and KF (solid) in four basins, in centimeters of equivalent water height, over the full test window. The panels span the range of per-basin outcomes at a 1-month lead time: (a) and (b) two arid basins with pronounced month-to-month residual scatter and large filter gains; (c) a mid-latitude river basin where the gain is a few percent; (d) the Amazon, where damped persistence is marginally better. Each panel states the 1-month RMSE of both forecasts over the window.

time and slightly higher error than damped persistence at 6 months. The pooled lag ridges have lower error than KF at 4 and 5 months, although these differences are not significant.

On the JPL product, with 228 basins and the same folds, KF's margin over the stronger damped-persistence variant holds at 2 to 6 months ($+1.5$ to $+6.3\,\%$, $p < 0.02$), with no significant difference at 1 month ($+0.6\,\%$, $p = 0.60$). Against the per-basin lag ridge the pattern is the same: KF is behind at 1 month ($-0.66\,\%$, $p = 0.46$) and ahead by $+2.11$ to $+6.67\,\%$ at 2 to 6 months ($p < 0.03$; Table B2). Damped persistence itself has a smaller standardized 1-month error on JPL (0.67 against 1.08 on CSR), so the JPL residual is more persistent from one month to the next and leaves the filter less to remove at the shortest lead. Which processing choices produce this is not tested (Sect. 5.3).

4.2 Why the filter improves on damped persistence

We use two ablation experiments to identify the source of the margin (Fig. 4). The zero-r variant, the same model refit with $r = 0$, loses to the full filter at every lead time, by $5.35\,\%$ at 1 month, rising to $12.65\,\%$ at 5 months ($p < 0.001$ at every lead time). Scored against the regression variant at every lead time, the zero-r variant is $+0.90\,\%$ ahead at 1 month and 0.93 to $11.57\,\%$ behind at 2 to 6 months; against the ρ^h variant, the stronger one at 1 month, it is $0.40\,\%$ behind. At a 1-month lead time the AR(1) structure without the model-noise term is therefore within 1 % of damped persistence in either direction. The corresponding mean-squared-error ratios multiply: $0.9465 \times 1.0040 \simeq 0.9502$, recovering the $4.98\,\%$ improvement in Table 2. Allowing a nonzero model-noise variance accounts for the improvement within this model family. Because r is estimated from the residual under the state-space assumptions (Sect. 3.4), the ablation attributes the margin to filtering within the model class rather than to a particular physical noise source.

The second ablation asks whether the two-variance variant (Sect. 3.4), with a separate model-noise variance for GRACE-FO months, improves the forecast. It is worse than the single-variance filter at every lead time: $-1.84\,\%$ at 1 month ($p < 0.001$) and -0.80 to $-0.38\,\%$ at 2 to 6 months ($p < 0.01$). Per basin at a 1-month lead time, after false-discovery-rate control, none of the 234 basins has a significant improvement or deterioration. The pooled difference is therefore not accompanied by individually significant basin differences. The first fold's training window ends in May 2019 and holds fewer than 12 GRACE-FO months, so all 234 of its basin-fold fits fall back to the single variance (Sect. 3.4). Every scored month is a GRACE-FO month. The separate GRACE-FO variance is estimated from fewer than seven years in any fold, which may limit its precision. The single-variance filter has lower test error and is retained for the remaining comparisons.

4.3 Forecast models built on the filter

Table 3 scores the own-basin models against the filter. KF-R1, which adds one regression coefficient on the current filtered state, is weaker than KF at 1 and 2 months, by -0.17 and $-1.01\,\%$, and stronger at 3 to 6 months, by $+0.44$ to $+2.66\,\%$ ($p < 0.001$). Both use the same current filtered state and require only the storage record. KF-R12, which uses twelve months of the filtered state instead of one, improves on KF at every lead time, by $+0.63$ to $+1.81\,\%$ ($p < 0.02$). KF-R12E adds the twelve-month forcing window and gains $+7.65\,\%$ over KF at 1 month (95 % interval $+4.7$ to $+10.1\,\%$), a gain that falls from $+5.39\,\%$ at 2 months to $+2.09\,\%$ at 5 months ($p < 0.001$), and a non-significant $+0.87\,\%$ at 6 months ($p = 0.12$). Relative to

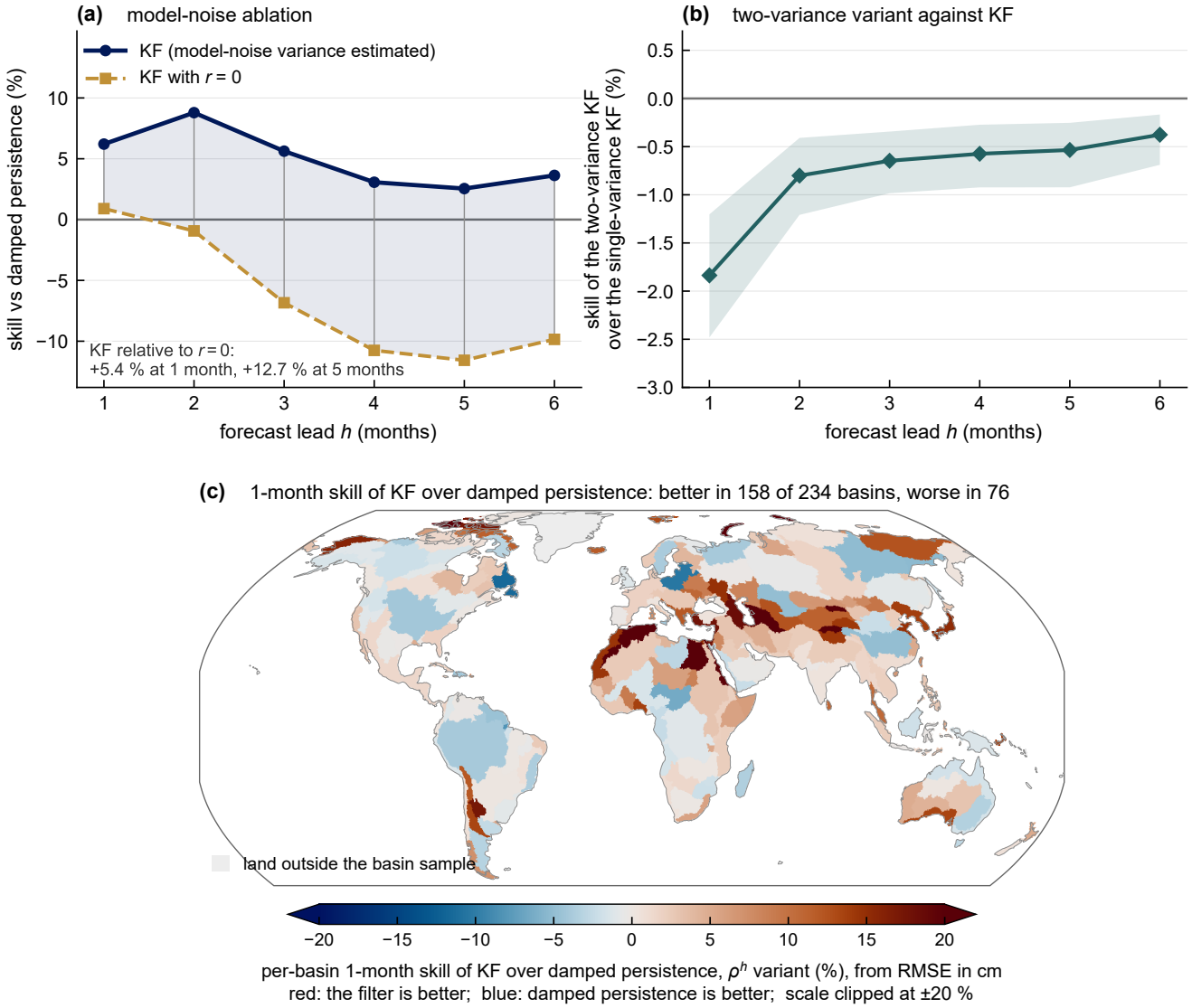


Figure 4. Attribution of the filter's margin. (a) Skill (%) against the regression damped-persistence variant at every lead time for KF and for the zero-r variant (KF with $r = 0$). Shading shows the difference in skill relative to this common reference, without a confidence interval. The annotated improvement of KF relative to $r = 0$ uses the latter's MSE as its denominator and therefore differs from the shaded vertical gap. Because the reference here is the regression variant at every lead time, KF has a skill of $+6.21\,\%$ at a 1-month lead time rather than the $+5.0\,\%$ of Table 2, which uses the stronger ρ^h variant at that lead time. (b) The two-variance variant against the single-variance KF by lead time, with moving-block bootstrap 95 % confidence intervals; negative means the split hurts. (c) Per-basin 1-month skill of KF over the ρ^h damped-persistence variant on the 234 basins, computed from RMSE in centimeters; KF is better in 158 basins and damped persistence in 76. The color scale is clipped at $\pm 20\,\%$, and land outside the basin sample is pale grey. Basin partition based on HydroSHEDS (Lehner and Grill, 2013); base map made with Natural Earth (public domain).

damped persistence this is $+12.2\,\%$ at 1 month and $+13.7\,\%$ at 2 months (Table 2), the largest margins in the study. KF-R12E is the best own-basin model in standardized units at 1 to 4 months. At 5 and 6 months, KF-R1, with one coefficient on the current filtered state, is better by less than 1 % (Appendix B). The forcing window improves on KF-R1 only up to a lead time of 4 months.

Table 3. Own-basin models against the Kalman filter: pooled skill (%) against KF at each lead time h (months) on the same basin-months (upper value), with the Diebold–Mariano p value below it. Positive means the model improves on the filter; the best value in each column is in bold.

Forecast	h=1	h=2	h=3	h=4	h=5	h=6
KF-R1	-0.17	-1.01	$+0.44$	$+2.36$	$\mathbf{+2.66}$	$\mathbf{+1.59}$
	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
KF-R12	$+0.81$	$+1.48$	$+1.76$	$+1.81$	$+1.49$	$+0.63$
	0.0035	< 0.001	< 0.001	< 0.001	< 0.001	0.015
KF-R12E	$\mathbf{+7.65}$	$\mathbf{+5.39}$	$\mathbf{+4.22}$	$\mathbf{+3.19}$	$+2.09$	$+0.87$
	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	0.12

The KF-R1 p values are two-sided tests of a difference that is negative at $h = 1$ and 2 and positive at $h = 3$ to 6.

The gain from the forcing window is not uniform across basins. Figure 5 maps the per-basin 1-month skill of KF-R12E over KF and plots it against each basin's median training-window storage standard deviation across folds. The relation is weak across the sample (Spearman $\rho = -0.14$). The 45 basins that are significant under false-discovery-rate control, 44 helped and 1 hurt, all have training-window standard deviations below 9 cm, and none of the highest-variance basins gains significantly. One standardized unit represents a smaller physical anomaly in a low-variance basin. Improvements there contribute more to the standardized comparison than to pooled error in centimeters. This scaling makes the ranking of the models sensitive to how basins are weighted (Sect. 4.5).

None of the nonlinear models improves on KF-R12E given the same twelve months of history (Table 4 and Fig. 6). At 1 to 3 months KF-R12E has the lowest standardized RMSE of every model and seed. The MLP, which sees the identical feature vector, is significantly worse in every seed at every lead time ($p < 0.001$), and the residual MLP and both LSTM seeds fall between KF-R12E and KF-R1E. Because the LSTM holds out the last 15 % of its training months for early stopping, its comparison with a ridge fit on all training months is not matched on training data. We therefore refit KF-R12E on the LSTM's training months. Against the two-seed LSTM ensemble the refit ridge is ahead by $+1.02\,\%$ at 1 month ($p = 0.094$), $+2.76\,\%$ at 2 months and $+2.27\,\%$ at 3 months (both $p < 0.001$). Its difference from the ridge fit on all months is not significant (differences within $0.2\,\%$, $p \geq 0.19$). Under the architectures and training budget tested on CSR, the nonlinear models provide no improvement over this linear representation of the storage and forcing history.

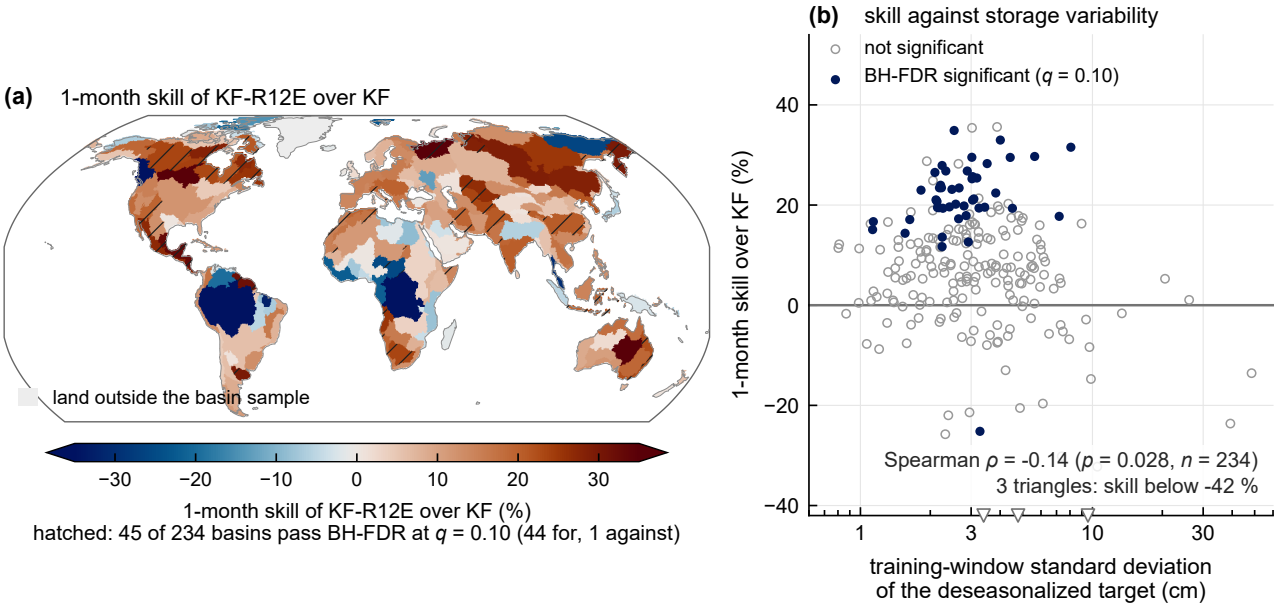


Figure 5. Where the forcing window helps. (a) Per-basin 1-month skill (%) of KF-R12E over KF on the 234 basins; basins significant under Benjamini–Hochberg false-discovery-rate control at $q = 0.10$ are hatched, and land outside the basin sample is pale grey. (b) The same skill against each basin's median training-window storage standard deviation across folds, in centimeters of equivalent water height; filled markers are the 45 significant basins (44 helped, 1 hurt). Three basins fall below the lower axis limit of $-42\,\%$ and are marked by downward triangles at their actual storage standard deviations. The Spearman correlation is -0.14. The color scale in (a) is clipped at $\pm 35\,\%$. Basin partition based on HydroSHEDS (Lehner and Grill, 2013); base map made with Natural Earth (public domain).

Table 4. Nonlinear models against KF-R12E: standardized pooled RMSE at lead times of 1 to 3 months on the same basin-months (n as in Sect. 3.2). Lower is better. Seeds are listed separately; the LSTM ensemble is scored in the text through the comparison matched on training months.

Forecast	h=1	h=2	h=3
KF	1.0490	1.1954	1.2734
KF-R1E	1.0254	1.1930	1.2685
KF-R12	1.0448	1.1865	1.2622
KF-R12E	**1.0081**	**1.1627**	**1.2462**
LSTM, seed 0 / seed 1	1.0132 / 1.0188	1.1758 / 1.1851	1.2578 / 1.2630
MLP, seed 0 / seed 1	1.0235 / 1.0291	1.1783 / 1.1935	1.2602 / 1.2799
Residual MLP, seeds 0 / 1 / 2	1.0148 / 1.0156 / 1.0145	1.1760 / 1.1788 / 1.1773	1.2635 / 1.2530 / 1.2540

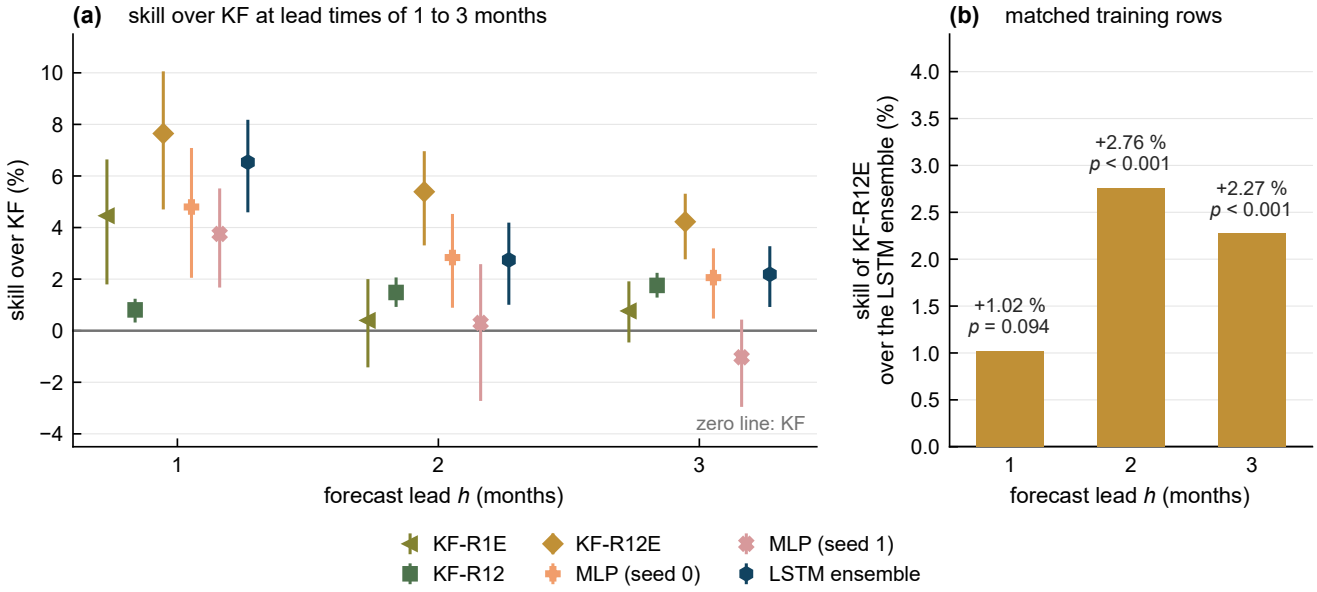


Figure 6. Nonlinear models against KF-R12E. (a) Skill (%) over KF at lead times of 1 to 3 months for KF-R1E, KF-R12, KF-R12E, the two seeds of the MLP and the two-seed LSTM ensemble, with moving-block bootstrap 95 % confidence intervals. The residual MLP appears in Table 4 only. (b) Skill of KF-R12E over the LSTM ensemble when the ridge is refit on the LSTM's training months, with the Diebold–Mariano p value.

4.4 Comparison with GRACE-FCast on CSR and JPL

Table 5 and Fig. 7 compare our forecasts with the individual GRACE-FCast members on the fully covered basins, in standardized units. On CSR, over 209 basins and 60 months (12,540 basin-months per lead time), KF, with no forcing input, has lower error than the full product at a 1-month lead time, by $16.4\,\%$ (95 % interval $+4.9$ to $+26.6\,\%$). The difference is not significant at 2 months; KF has higher error from 3 months on, by 16.3 to $35.4\,\%$ ($p < 0.002$). Against the non-seasonal component, which removes the seasonal and trend re-extrapolation, KF has lower error at 1 and 2 months ($+28.4$ and $+7.5\,\%$), no significant difference at 3 months, and higher error from 4 months on. KF-R12E follows the same pattern of significant differences against both product variants, with larger 1-month margins ($+21.5$ and $+32.8\,\%$) and interpolated zero crossings slightly later (Fig. 7). On JPL, over 67 basins and 59 months (3,953 basin-months per lead time), only KF is compared with GRACE-FCast. Against the full product, KF has significantly lower error at 1 month, no significant difference at 2 and 3 months, and higher error from 4 months on. Against the non-seasonal component, KF has significantly lower error at 1 and 2 months, no significant difference at 3 and 4 months, and higher error from 5 months on. Its 1-month improvements are $+43.7\,\%$ against the full product and $+52.9\,\%$ against the non-seasonal component, in which case KF's mean squared error is less than half the product's.

Table 5. Comparison with GRACE-FCast on the fully covered basins. Skill (%) of our forecast over GRACE-FCast at each lead time h (months), upper value, with the Diebold–Mariano p value below it; positive means our forecast has the lower error. CSR: 209 basins, 60 months, 12,540 basin-months per lead time. JPL: 67 basins, 59 months, 3,953 basin-months per lead time, same convention. The last two rows count the CSR basins, of 209, in which the full product has the lower error at $h = 1$ and $h = 6$.

Pair	h=1	h=2	h=3	h=4	h=5	h=6
CSR						
KF vs. full	$+16.4$	-6.0	-16.3	-23.2	-28.9	-35.4
	0.0022	0.23	0.0014	< 0.001	< 0.001	< 0.001
KF vs. non-seasonal	$+28.4$	$+7.5$	-4.0	-13.4	-20.3	-27.1
	< 0.001	0.018	0.13	< 0.001	< 0.001	< 0.001
KF-R12E vs. full	$+21.5$	-1.8	-12.1	-19.6	-26.4	-34.6
	< 0.001	0.74	0.032	0.0017	< 0.001	< 0.001
KF-R12E vs. non-seasonal	$+32.8$	$+11.2$	-0.2	-10.1	-18.0	-26.4
	< 0.001	0.0021	0.94	0.0032	< 0.001	< 0.001
Damped persistence vs. full	$+11.9$	-16.9	-30.9	-42.2	-47.9	-54.3
	0.029	0.0011	< 0.001	< 0.001	< 0.001	< 0.001
JPL						
KF vs. full	$+43.7$	$+9.8$	-6.2	-19.1	-32.1	-41.9
	< 0.001	0.087	0.31	0.0017	< 0.001	< 0.001
KF vs. non-seasonal	$+52.9$	$+22.2$	$+7.0$	-6.2	-18.0	-27.9
	< 0.001	< 0.001	0.15	0.16	< 0.001	< 0.001
Damped persistence vs. full	$+43.2$	$+4.1$	-19.0	-39.2	-53.1	-65.3
	< 0.001	0.49	0.0047	< 0.001	< 0.001	< 0.001
CSR basin counts (of 209)						
Basins, full product has lower error than KF	98					157
Basins, full product has lower error than KF-R12E	87					162

JPL values are converted from the product-first skill s of the deposited tables as $1 - 1/(1 - s)$, the reciprocal of the mean-squared-error ratio.

At the level of individual basins, the 1-month result is close to an even split, and the number of basins in which GRACE-FCast is better increases with lead time. On CSR the full product has the lower error in 98 of the 209 basins against KF at 1 month and in 157 at 6 months, and in 87 against KF-R12E at 1 month and 162 at 6 months. The pooled 1-month margin therefore reflects large gains in a subset of basins and not a consistent advantage across basins.

The short-lead comparison is sensitive to the available inputs. GRACE-FCast's non-seasonal networks use forcing predictors without lagged storage, while its full forecast extrapolates seasonal and trend terms from past GRACE observations. Damped persistence, KF and its corrections use the recent residual directly. Damped persistence has lower error than the full product at 1 month, by $+11.9\,\%$ on CSR and $+43.2\,\%$ on JPL, and loses to it from 2 months on CSR and from 3 months on JPL (Table 5). The comparison with damped persistence more directly assesses the contribution of filtering. On the full benchmark sample, the 1-month improvement is $+5.0\,\%$ on CSR (Table 2), with no significant difference on JPL ($+0.6\,\%$, $p = 0.60$; Sect. 4.1). The 1-month margin over the product is much larger on JPL than on CSR. The fully covered JPL set contains 67 basins selected by native-cell containment, and the JPL member of GRACE-FCast was trained on an earlier release than the one we score it against (Sect. 2.4). We cannot separate these two effects.

4.5 Sensitivity to basin weighting

The comparisons above use basin-standardized errors. In pooled centimeter errors, high-variance basins contribute more to squared error. The five forecasts reported in that unit differ by about half a centimeter at most, and the differences among the filter-based models are untested (Fig. 2b; Appendix C). KF-R12 has the lowest value at 1 to 3 months and KF-R1 at 4 to 6 months, and KF-R12E, the best model in standardized units, is 0.1 cm behind the filter at 1 month. Damped persistence has the highest RMSE at every lead time in this weighting as well. The filter's margin over persistence therefore does not depend on the weighting, but the gain from the forcing window does. It is concentrated in the low-variance basins of Fig. 5, which receive full weight in the standardized metric and little weight in the centimeter metric. Any ranking of models on this target should therefore state the weighting used. In this paper, "best own-basin model" means lowest pooled standardized error.

4.6 Neighboring-basin predictors

Spatial and graph-based TWSA methods use information from other locations: Li and Kusche (2026) shift forcing predictors across space, Arzoumanidis et al. (2026) build a proximity and correlation basin graph for reconstruction, and Steidl and Zhu (2025) forecast with a hierarchical basin graph. We tested a storage-based version of this idea: whether one other basin's filtered state, propagated to the target month and added as a feature to KF-R1, improves the forecast. On CSR a correlation-selected neighbor was worth $+0.31\,\%$ over KF-R1 at 1 month, ahead of all 50 seed-matched random neighbor graphs and all 99 amplitude-adjusted Fourier-transform surrogates (Schreiber and Schmitz, 1996), and nothing at 2 to 6 months. None of this is reproduced on JPL. There, every neighbor variant is worse than KF-R1 at every lead time. The correlation-selected neighbor beats none of the 50 random graphs at any lead time, the distance-selected neighbor beats at most 26 of 50, and the real graph beats none of the 99 surrogates. The reversal across processing centers provides no support for adopting this neighboring-basin correction as a general improvement.

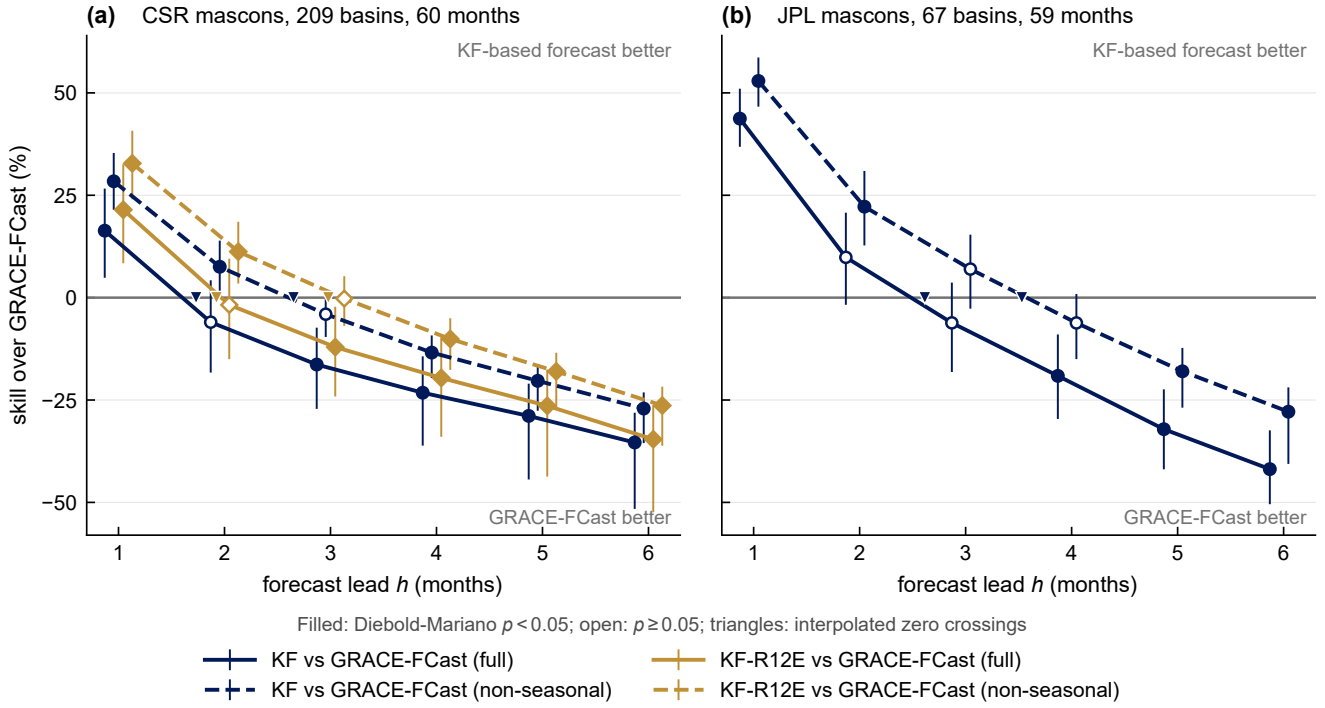


Figure 7. Comparison with GRACE-FCast on the fully covered basins. (a) CSR, 209 basins and 60 months: skill (%) of KF and of KF-R12E over GRACE-FCast, full and non-seasonal, by lead time, with moving-block bootstrap 95 % confidence intervals; positive means our forecast has the lower error. (b) JPL, 67 basins and 59 months, KF only. Filled markers are significant at $p < 0.05$, open markers are not, and the triangles on the zero line mark the interpolated lead time at which each curve crosses zero. Damped persistence, not drawn, also has lower error than the full product at a 1-month lead time on both products (Table 5); the 1-month advantage is therefore also present without Kalman filtering.

5 Discussion

5.1 What the benchmark means for reported skill

Filtering improves the use of recent storage history as a forecast benchmark. In this evaluation, a three-parameter filter has lower error than both persistence variants on CSR and at most tested leads on JPL (Table 2; Appendix B). A model's improvement over climatology or raw persistence therefore does not establish an improvement over the information available from the storage record. The size of this difference depends on the target, product, and lead time. The percentages reported here cannot be added across reference forecasts or used to adjust skill reported in other studies without evaluating their predictions on a common target.

In physical units the margin is modest. At a 1-month lead time the pooled RMSE falls from 5.32 to 5.13 cm of equivalent water height and the per-basin median from 2.74 to 2.66 cm (Appendix C). The gain is concentrated in small and arid basins

(Sect. 4.1). These are the basins in which GRACE errors are largest relative to the signal: bias and leakage errors grow as basin size decreases (Longuevergne et al., 2010; Vishwakarma et al., 2018), and Li and Kusche (2026) report their lowest relative forecast skill in arid regions, where storage variability is small. The filter's gains follow the same pattern; whether the variation it removes there is observation error or fast hydrology is not resolved by the model (Sect. 3.4). In large humid basins such as the Amazon, the filter and damped persistence are nearly the same forecast (Fig. 3d). The benchmark therefore changes the accounting of learned skill more than it changes the error of the basins that dominate centimeter metrics.

GRACE-FCast has lower error at longer lead times in the external comparison. On the fully covered CSR sample, KF and KF-R12E have 10 to 35 % higher mean squared error than the full or non-seasonal GRACE-FCast forecast at 4 to 6 months. At 1 month, KF, KF-R12E and damped persistence have lower error than the full product; at 2 months, KF also has lower error than its non-seasonal component. These comparisons require an explicit reference forecast and cannot be inferred from full-signal correlation alone. Li and Kusche (2026) report a related sensitivity to the verification target: they find correlation above 0.8 over about 72 % of global land at a 1-month lead time for the full signal but over only 20 % once the seasonal cycle and trend are removed. They attribute the difference to the extrapolated components. Appendix C likewise shows higher correlations after the extrapolated seasonal cycle and trend are restored to our forecasts. Reporting full-signal metrics alongside deseasonalized skill makes clear whether a model improves the less predictable residual component.

Our result complements that of Nie et al. (2025) (Sect. 1), who compare architectures with lagged storage excluded. Here, a linear correction using filtered storage and forcing has lower CSR error than the nonlinear models tested on the same inputs (Sect. 4.3). The training-matched comparison retains this advantage over the LSTM at 2 and 3 months, but the 1-month difference is not significant. This supports retaining linear models as controls in architecture comparisons. It does not establish their superiority under other training budgets or on other targets. In particular, simulated, reconstructed and gap-filled targets, including HydroGlobe (Nie et al., 2025) and the products of Humphrey and Gudmundsson (2019), Yin et al. (2023) and Mo et al. (2022), have error structures that differ from direct mascon observations. The benefit of filtering those targets would need to be evaluated separately.

5.2 Initial conditions versus forcing

Seasonal hydrological forecasting experiments distinguish predictability from initial conditions and meteorological forcing. Initial conditions often contribute most at short leads, with forcing becoming more important as lead time increases (Wood and Lettenmaier, 2008; Shukla and Lettenmaier, 2011; Koster et al., 2010; Yossef et al., 2013; Pechlivanidis et al., 2020). Those results were established for streamflow and soil moisture in ensemble streamflow prediction experiments. The comparison in Sect. 4.4 is consistent with this pattern: KF has lower error at short leads, whereas GRACE-FCast has significantly lower error from 3 to 5 months onward, depending on the product and component (Table 5). The 1-month advantage also occurs for damped persistence. The crossover describes these forecast systems; it does not decompose skill into contributions from storage and forcing. Architecture, training data, gap treatment, predictor sets and target decomposition differ together in this comparison.

The own-basin ERA5-Land history improves the CSR linear correction through 4 months relative to KF-R1 (Sect. 4.3), while the external product retains its advantage over KF through the longest lead evaluated. These results motivate combining storage-based and forcing-based forecasts. A switching lead or blending weight would need to be selected and evaluated on separate data; the crossings in Fig. 7 do not by themselves establish such a rule.

5.3 Limitations

None of our models uses forecast forcing; the ERA5-Land window ends at the issue month. The comparisons therefore concern predictability from antecedent observations and reanalysis. They do not assess the benefit of adding meteorological forecasts.

Observation latency changes the operational meaning of these lead times. GRACE-FO solutions are released two to three months after the observation month (Mo et al., 2025). Predicting next month's storage from an observation released two to three months late requires an observation-relative lead of approximately 3 to 4 months. On CSR, those leads are at or beyond the crossover with GRACE-FCast. The common issue-date convention used here supports retrospective comparison; it does not reproduce the availability of all inputs in an operational forecast.

The cross-product evidence reported here concerns the benchmark forecasts and their comparison with GRACE-FCast. The detailed correction-model ranking in Sect. 4.3 is for CSR and should not be generalized to JPL. The fully covered JPL set contains 67 basins selected by native-cell containment, and the JPL member of GRACE-FCast was trained on an earlier release than we score it against. For that reason the size of the 1-month margin there should not be compared with CSR's. In addition, the filter's 1-month margin over damped persistence does not replicate on JPL: no significant difference is detected at 1 month, while KF improves on damped persistence at 2 to 6 months (Sect. 4.1). The smaller JPL persistence error is consistent with less unpredictable month-to-month variation, but the contribution of individual processing choices was not isolated.

All results are for basin averages. Most published data-driven TWSA forecasts are gridded (Li et al., 2020, 2024; Li and Kusche, 2026). Spatial aggregation changes both the signal and its error structure, so the filter's performance at grid scale remains to be tested, with (ρ, q, r) estimated for each cell.

The filter's definition across the 2017–2018 gap follows from its prediction step, which runs whenever an update is skipped. However, no scored forecast is issued across the gap (Sect. 3.2), so this property is not tested here.

The model-noise term is not an estimate of instrument noise: in 259 of 1170 basin-fold fits the likelihood sets r to effectively zero (Sect. 3.4). The ablation attributes the filter's margin to filtering within this model class. What is filtered out is presumed to be a mixture of physical variability and the errors listed in Sect. 3.4, and we do not estimate their shares. The results concern forecast error against the observed residual, not the accuracy of the satellite data.

The ranking of models above the filter depends on basin weighting. The forcing window's gain is concentrated in low-variance basins that contribute less to pooled centimeter error (Sect. 4.5), while the filter's margin over persistence holds in both weightings. The comparison with GRACE-FCast is not symmetric. Their training uses the gap reconstruction of Yin et al. (2023), whereas our models are fitted to the observed series and this study's target definition. The 1-month advantage also occurs for damped persistence. The external comparison excludes fold 5 and basins without adequate forecast-grid coverage.

Finally, the evidence rests on two mascon products of one set of observations, five test folds all in the post-2019 climate, and per-basin estimates based on 83 months at a 1-month lead time. The LSTM, MLP and residual MLP were trained with the fixed architectures of Sect. 3.5, early stopping and no hyperparameter search, with two seeds each for the first two and three for the third. Table 4 reports individual seeds, and the training-matched comparison uses the LSTM seed ensemble. Under this budget, none has lower CSR error than KF-R12E; whether further tuning changes this ordering remains untested. The short record also limits inference about small differences. Our uncertainty estimates account for overlapping forecasts, but sensitivity to longer HAC bandwidths and bootstrap blocks has not been evaluated.

6 Conclusions

For deseasonalized basin-average GRACE storage, the per-basin Kalman forecast improves on the stronger damped-persistence variant at all six tested leads on CSR and at 2 to 6 months on JPL. Refitting without a model-noise term removes the CSR 1-month advantage, supporting the use of temporal filtering within this model family. We recommend reporting skill against this benchmark alongside persistence and physical-unit errors.

A linear correction using twelve months of filtered storage and ERA5-Land history has the lowest standardized CSR error at short leads among the correction models tested. Its improvement is concentrated in low-variance basins and does not extend to pooled error in centimeters. In the comparisons with individual GRACE-FCast members on the fully covered basins, GRACE-FCast has lower standardized error at longer leads, whereas both KF and damped persistence have lower error at 1 month. Any evaluation of storage forecasts, including any future combination of storage-based and forcing-based predictions, has to account for this dependence on lead time, error scaling and input availability.

Appendix A: Model names and repository identifiers

The result files in the data repository identify each model by the identifier used in the analysis code. Table A1 maps the names of Table 1 to those identifiers, which appear nowhere else in the paper. Seed suffixes (`_s0`, `_s1`, `_s2`) denote single training seeds and `_ens` the seed ensemble.

Table A1. Model names used in the paper and the corresponding identifiers in the analysis result files.

Paper name	Identifier(s)
Climatology	`climatology_zero`
Persistence	`persistence`
Damped persistence, ρ^h and regression variants	`damped_persistence_rho`, `damped_persistence_reg`
Pooled lag ridge, and index variant	`ridge_own_lags`, `ridge_own_plus_indices`
Per-basin lag ridge	`ridge_own_perbasin`
KF	`kalman_ar1`
KF, $r = 0$	`kalman_r0`
KF, two variances	`kalman_mission`
KF-R1	`kalman_own_ridge` (`ridge_own` in the sequence-model files)
KF-R12	`ridge_own_flat12`
KF-R12E	`ridge_own_era5_flat12`; refit on the LSTM's training months `ridge_own_era5_flat12_train85`
KF-R1E	`ridge_own_era5`
LSTM	`lstm_own_era5_s0`, `lstm_own_era5_s1`, `lstm_own_era5_ens`
MLP	`mlp_own_era5_flat12_s0`, `mlp_own_era5_flat12_s1`
Residual MLP	`resmlp_own_era5_s0`, `resmlp_own_era5_s1`, `resmlp_own_era5_s2`
Neighbor ridge, correlation- and distance-selected	`kalman_corr_top1`, `kalman_geo_top1`
GRACE-FCast, full and non-seasonal	`li_lstm_full`, `li_lstm_nonseas`
Fully covered basin set	`joint_full_cells`; matched set `all_matched`

Appendix B: The full benchmark tables

Table B1 gives, for every forecast in Table 2, the pooled standardized RMSE and the Diebold–Mariano (DM) p value against the stronger damped-persistence variant at all six lead times on the CSR product. The adjacent rows are labelled explicitly as RMSE and two-sided DM p so that each pair is read as one forecast result. Table B2 gives the corresponding skills and p values for the JPL benchmark set: the benchmark forecasts, KF-R1 and the lag ridges. The twelve-month correction models are reported for CSR only. A small p value indicates a difference from the reference in either direction; it does not by itself indicate improvement. The pooled tests are unadjusted across models and horizons; Benjamini–Hochberg adjustment is used only for per-basin comparisons.

Table B1. CSR product: pooled standardized RMSE and two-sided Diebold–Mariano (DM) p value against the stronger damped-persistence variant for every forecast of Table 2 at lead times h of 1 to 6 months, over 234 basins and five folds. The pooled test uses the monthly mean loss difference across basins as its temporal observations (83 at $h = 1$), not the 19,422 scored basin-months as independent replicates. A small p value indicates a difference in either direction, not necessarily an improvement. The reference row has no test.

Forecast	Metric	h=1	h=2	h=3	h=4	h=5	h=6
Climatology	RMSE	1.6163	1.6354	1.6525	1.6698	1.6881	1.7055
	p	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Persistence	RMSE	1.1825	1.3863	1.4442	1.4837	1.5224	1.5877
	p	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Damped persistence, weaker variant	RMSE	1.0831	1.2552	1.3519	1.4354	1.4953	1.5478
	p	< 0.001	0.13	< 0.001	< 0.001	< 0.001	< 0.001
Damped persistence, stronger variant	RMSE	1.0761	1.2516	1.3108	1.3573	1.4132	1.4754
	p	–	–	–	–	–	–
Pooled lag ridge	RMSE	1.0698	1.2214	1.2770	1.3324	1.3906	1.4570
	p	0.15	< 0.001	< 0.001	< 0.001	0.0026	0.013
Pooled lag ridge, index variant	RMSE	1.0708	1.2230	1.2766	1.3316	1.3906	1.4578
	p	0.24	< 0.001	< 0.001	< 0.001	0.0026	0.015
Per-basin lag ridge	RMSE	1.0596	1.2139	1.2826	1.3433	1.4105	1.4776
	p	< 0.001	< 0.001	< 0.001	< 0.001	0.48	0.70
KF	RMSE	1.0490	1.1954	1.2734	1.3363	1.3951	1.4484
	p	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
KF-R1	RMSE	1.0499	1.2014	1.2706	1.3204	1.3764	1.4369
	p	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
KF-R12	RMSE	1.0448	1.1865	1.2622	1.3241	1.3846	1.4438
	p	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
KF-R12E	RMSE	1.0081	1.1627	1.2462	1.3148	1.3804	1.4421
	p	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001

Table B2. JPL product: pooled skill (%) against the stronger damped-persistence variant and two-sided Diebold–Mariano (DM) p value at lead times h of 1 to 6 months, over 228 basins and five folds ($n = 19{,}152$ scored basin-months at $h = 1$). Pooled p values use the monthly mean loss difference across basins as temporal observations, not basin-months as independent replicates. The reference is the ρ^h variant at $h = 1$ and 2 and the regression variant at $h = 3$ to 6. A small p value indicates a difference in either direction, not necessarily an improvement. The reference row has no test.

Forecast	Metric	h=1	h=2	h=3	h=4	h=5	h=6
Climatology	Skill (%)	-374.3	-184.4	-141.3	-119.3	-95.0	-75.5
	p	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Persistence	Skill (%)	-4.4	-7.1	-4.7	-3.3	-2.1	$+0.4$
	p	0.021	0.0042	0.050	0.21	0.45	0.88
Damped persistence, weaker variant	Skill (%)	-0.9	-1.3	-6.9	-15.0	-14.0	-12.2
	p	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Damped persistence, stronger variant	Skill (%)	0.0	0.0	0.0	0.0	0.0	0.0
	p	–	–	–	–	–	–
Pooled lag ridge	Skill (%)	$+0.5$	$+1.7$	$+2.2$	-0.5	-1.1	-0.9
	p	0.62	0.18	0.058	0.57	0.18	0.31
Pooled lag ridge, index variant	Skill (%)	$+0.7$	$+2.0$	$+2.8$	$+0.1$	-0.5	-0.6
	p	0.47	0.17	0.042	0.90	0.63	0.62
Per-basin lag ridge	Skill (%)	$+1.2$	$+3.2$	$+2.3$	-0.6	-1.6	-2.1
	p	0.33	0.0059	0.0087	0.43	0.081	0.058
KF	Skill (%)	$+0.6$	$+6.3$	$+4.6$	$+1.5$	$+2.4$	$+4.7$
	p	0.60	< 0.001	< 0.001	0.017	< 0.001	< 0.001
KF-R1	Skill (%)	$+0.3$	$+5.2$	$+5.1$	$+3.3$	$+3.5$	$+4.2$
	p	0.80	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001

Appendix C: Conventional metrics in centimeters

Table C1 reports physical-unit errors, correlation and Nash–Sutcliffe efficiency for the five forecasts in Sect. 4.5. Full-signal forecasts restore each fold's frozen training climatology. Because forecast and observation are shifted by the same climatology, the centimeter errors are unchanged, but correlation and efficiency both rise. For example, at a 1-month lead, damped persistence has a median full-signal correlation of 0.86 and a median full-signal efficiency of 0.72, against 0.62 and 0.33 for the same forecast scored on the residuals alone. The extrapolated seasonal cycle and trend account for the difference.

Table C1. Conventional metrics for the five forecasts reported in centimeters: pooled RMSE and per-basin median RMSE in centimeters of equivalent water height, and per-basin median correlation (CC) and Nash–Sutcliffe efficiency (NSE) on the residual and on the full signal, over 234 basins, at lead times h of 1 to 6 months. The damped-persistence rows use the stronger variant at each lead time.

Forecast	h	Pooled RMSE (cm)	Median RMSE (cm)	CC anom.	CC full	NSE anom.	NSE full
Damped persistence	1	5.321	2.74	0.62	0.86	0.33	0.72
	2	6.156	3.32	0.46	0.81	0.09	0.59
	3	6.383	3.57	0.39	0.79	0.02	0.54
	4	6.447	3.65	0.34	0.78	−0.01	0.51
	5	6.689	3.78	0.30	0.77	−0.04	0.48
	6	6.935	3.96	0.23	0.75	−0.08	0.43
KF	1	5.128	2.66	0.64	0.86	0.36	0.74
	2	5.650	3.23	0.50	0.82	0.18	0.63
	3	5.949	3.46	0.43	0.80	0.06	0.56
	4	6.161	3.57	0.37	0.78	0.00	0.52
	5	6.413	3.72	0.30	0.76	−0.02	0.48
	6	6.690	3.84	0.24	0.74	−0.05	0.44
KF-R1	1	5.133	2.66	0.64	0.86	0.36	0.74
	2	5.691	3.26	0.50	0.82	0.17	0.63
	3	5.931	3.46	0.43	0.80	0.06	0.56
	4	6.064	3.55	0.37	0.78	0.01	0.53
	5	6.288	3.68	0.31	0.76	−0.01	0.49
	6	6.603	3.82	0.25	0.74	−0.05	0.45
KF-R12	1	5.118	2.66	0.64	0.86	0.36	0.74
	2	5.634	3.19	0.51	0.82	0.18	0.63
	3	5.915	3.39	0.45	0.81	0.08	0.57
	4	6.107	3.52	0.37	0.78	0.03	0.53
	5	6.366	3.72	0.30	0.76	−0.02	0.49
	6	6.678	3.85	0.26	0.74	−0.05	0.44
KF-R12E	1	5.231	2.56	0.69	0.88	0.42	0.76
	2	5.741	3.12	0.55	0.83	0.22	0.65
	3	6.002	3.37	0.48	0.81	0.10	0.58
	4	6.163	3.48	0.39	0.78	0.04	0.54
	5	6.463	3.69	0.33	0.76	−0.02	0.48
	6	6.774	3.87	0.27	0.74	−0.05	0.43

Code and data availability. CSR GRACE and GRACE-FO mascon solutions are available from the Center for Space Research (https://www2.csr.utexas.edu/grace/) and JPL mascon solutions from NASA PO.DAAC (https://podaac.jpl.nasa.gov/). ERA5-Land data are available from the Copernicus Climate Data Store (https://cds.climate.copernicus.eu/) (Muñoz-Sabater et al., 2021). Climate indices are available from NOAA PSL (https://psl.noaa.gov/data/climateindices/). The GRACE-FCast hindcast is available from PANGAEA (Li, 2025). The analysis code is at https://github.com/ishaankejriwal/graceseespaper. The benchmark forecasts, the per-prediction files and every result table used in this paper, together with a manifest mapping each table and figure to its source file, are being prepared for archival deposit before submission.

Author contributions. I.K. designed the study, wrote the analysis code, performed the experiments, and wrote the manuscript. I.W. validated the study, wrote the manuscript, adapted the experiment to fit JPL mascons, and replicated the experiments using JPL mascons. I.K. and I.W. share first authorship.

Competing interests. The authors declare that they have no conflict of interest.

Acknowledgements. The authors thank the Center for Space Research (University of Texas at Austin) and the Jet Propulsion Laboratory for the open mascon solutions and ancillary files, the Copernicus Climate Change Service for ERA5-Land, NOAA PSL for the climate indices, and Li and Kusche for publishing their gridded hindcast, which made the head-to-head comparison possible.

Financial support. This research received no external funding.

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